



WAREHOUSING & ARTIFICIAL INTELLIGENCE

By Ti Research



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WAREHOUSING & ARTIFICIAL INTELLIGENCE (AI)

As with every area within the supply chain and logistics landscape, warehouses, along with storage and fulfilment facilities, are impacted by the technology transformation now taking place. Warehouse operations have had to adjust to the shift in logistics patterns such as high velocity cross docking, Just in Time deliveries, kitting, return and repair, and a range of other activities rather than just being a 'shed'. Many of these activities have been supported by dedicated information systems that are integrated with other solutions such as RFID scanners, handling systems and pick, pack and despatch operations. But even with these capabilities, companies are now expecting, or expected to have, their warehouse facilities to operate within an extensive process network that can often span entire supply chains.

These demands go way beyond the traditional role of a warehouse. This continuous demand for flexible operations has huge implications for the information systems infrastructure. By design, systems have to be open and networked, with the implication that everything happening is 'visible' to the operating partners. The other significant factor that will impact all of the above, is the general availability of artificial intelligence (AI) and its use in augmenting or directing operations.

So existing systems are having to evolve into more open and interconnected platforms and the latest solutions are likely to be modular and collaborative, exploiting standards to operate as part of a federation of capabilities. By definition, many of these systems are provided as Cloud services (perhaps with a local instance running on a physical server in case of unexpected network failure) with high availability.

So what do these systems comprise and what are they likely to be doing?

- **The Rise of the Robots:** It's not just science fiction anymore. We're seeing a surge in autonomous mobile robots (AMRs) and collaborative robots (cobots). AMRs handle hauling tasks, while cobots assist with picking and packing, making warehouse operations faster and more accurate.
- **Modular and Flexible Solutions:** Gone are the days of giant, inflexible and expensive ERP based automation systems. Today's solutions are designed to be modular and scalable. This means you can add or remove components as your needs change, making automation more accessible for warehouses of all sizes.
- **The Power of the Internet of Things (IoT):** Imagine a warehouse where everything is connected and talking to each other. That's the power of IoT! Sensors on shelves, robots, and inventory can provide real-time data, allowing for better optimisation of space, resources, and labour.
- **Addressing the Labour Shortage:** The global labour shortage is a real challenge, and automation is helping to fill the gap. Robots can take over repetitive tasks, freeing up human workers for more complex jobs like problem-solving and customer service. Well that's the theory...
- **Smarter Warehouses:** We're seeing a rise in technologies like cloud-based warehouse

management systems (WMS), big data analytics, and even warehouse drones. These tools provide better visibility, improve decision-making, and allow for faster order fulfillment.

- **Sustainability in Focus:** Even warehouses are getting eco-friendly! There's a growing focus on using renewable energy sources and automation solutions that reduce waste and energy consumption.

AI is having a major impact on warehouse operations, making them more efficient and productive. Here are a few concrete examples:

- **Predictive Picking:** Imagine a system that can analyse past sales data and predict which items are most likely to be ordered next. AI-powered warehouse management systems can do exactly that, allowing warehouses to pre-pick and stage high-demand items, significantly reducing order fulfillment times.
- **Warehouse Layout Optimisation:** Warehouse space is valuable, and AI can help you use it more effectively. AI algorithms can analyse product dimensions, weight, and access frequency to create optimal storage layouts that minimise travel time for pickers and maximise storage capacity.
- **Smart Robots:** We mentioned robots before, but AI takes them to the next level. Warehouse robots equipped with AI can navigate warehouses autonomously, avoiding obstacles and even learning from their mistakes to improve efficiency over time. These robots can handle tasks like transporting pallets, picking items from shelves, or even packing orders.
- **Real-time Inventory Management:** Losing track of inventory can be a nightmare. AI-powered systems can use data from sensors on shelves and robots to provide real-time inventory visibility. This allows for more accurate stock control, reduces the risk of stockouts, and helps with forecasting future needs.
- **Preventative Maintenance:** Warehouse equipment breakdowns can disrupt operations significantly. AI can analyse sensor data from machinery to predict potential failures before they happen. This allows for preventative maintenance, reducing downtime and keeping the warehouse running smoothly.

Vision systems are a key component of the AI revolution within warehouses. They act as the eyes for robots and warehouse management systems, providing crucial visual data to optimise operations in several ways:

- **Enhanced Picking Accuracy:** Vision systems can identify and locate specific items, even when they're stacked haphazardly or partially obscured. This helps robots and pickers select the correct items, significantly reducing picking errors and improving order fulfillment accuracy.
- **Automated Quality Control:** Imagine a system that can automatically inspect incoming goods for damage or defects. Vision systems equipped with AI can do just that, analysing product images to

identify issues like dents, scratches, or missing parts. This ensures only high-quality products reach your customers.

- **Improved Damage Detection:** Warehouse environments can be rough on products. Vision systems can monitor conveyor belts and sorting areas, detecting damaged items before they get shipped out. This helps prevent customer dissatisfaction and potential returns.
- **Optimised Barcode Reading:** Traditional barcode scanners can struggle with wrinkled, dirty, or damaged labels. Vision systems can overcome these challenges, using advanced algorithms to read barcodes accurately and efficiently, ensuring smooth tracking and inventory management.
- **Advanced Robot Guidance:** Vision systems provide robots with real-time visual data about their surroundings. This allows robots to navigate warehouses autonomously, identify and pick items from shelves, and even place them in designated locations with greater precision and flexibility.

Overall, vision systems are the backbone of many AI-powered warehouse solutions. By providing crucial visual intelligence, they empower robots, improve decision-making, and contribute to a more efficient, accurate, and safe warehouse operation.

Here are some real-world examples of AI and vision systems working together in warehouses:

- **Amazon Robotics:** The e-commerce giant is a leader in warehouse automation. They utilise a network of robots equipped with AI and vision systems to navigate their massive fulfillment centers. These robots can pick and pack a wide variety of items, significantly improving order fulfillment speed and accuracy.
- **DHL's Sala360:** This innovative warehouse solution from DHL uses a combination of AI, robotics, and vision systems. Cameras and sensors track inventory in real-time, while robots autonomously retrieve and sort items. The system also uses AI for order fulfillment optimisation and predictive maintenance.
- **Walmart:** This retail giant is employing AI-powered systems to optimise picking routes within their warehouses. Cameras track inventory movement and worker activity, allowing the system to suggest the most efficient path for pickers to complete their tasks.
- **Bossa Nova Robotics:** This robotics company provides solutions specifically for warehouses. Their robots use AI and vision systems to identify and pick items from cluttered environments, making them ideal for warehouses with diverse inventory or where traditional automation struggles.
- **Honeywell Intelligrated:** This company offers a suite of warehouse automation solutions that integrate AI and vision systems. Their systems can handle tasks like automated palletising, where robots use vision to identify and stack items onto pallets for shipping.

In summary, modern warehouse operations are connected, integrated and agile environments where their owners have invested in the future. As the cost of technology continues to fall, investment strategies

should focus on finding and hiring the best people to manage in such environments. Collaboration is not just something that happens between partners, it will also occur between the workforce and smart systems.

For further analysis of the warehousing and contract logistics landscape, download Ti's new market report, *Global Contract Logistics 2024*. The report provides analysis of warehousing technology trends and industry expectations. The report also contains wider analysis of the contract logistics market including:

- 2023 & 2024 market sizing, growth rates and forecasts to 2028, split by region and market segment.
- Warehousing tracker data and analysis, including supply, demand, costs and outlook.
- State of Logistics 2024 survey analysis covering margins, volumes, and future investment trends.
- Global Logistics Contracts Analysis 2014-2024 covering logistics providers, client sectors, contract regions and contract length.
- Contract Logistics Procurement Survey 2023/2024 analysis.
- Competitive landscape analysis.
- In depth company profiles and Top 10 rankings.

For further information, or to download the report, visit ti-insight.com/product/global-contract-logistics



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